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Vapor Stone Rail Systems

Generic Software Documentation Project

Project # 7260

Software Design Description (SDD)
Real-Time Clock

Contract Number	-	CDRL Number	-
		Customer Ref. Number	-

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Revision List

Rev.	ECD#	Date	By	Description of changes
1	---	2011-09-14	Thomas Marmeth	Released with revision 1
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Preface

This Software Design Description (SDD) complies with the requirements of the IEEE 1016-1998 and IEEE Std 1558-2005, IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems approved on 2005-05-23.

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1 INTRODUCTION

This document is the Software Design Description (SDD) of the Real-Time Clock Module developed by Vapor Stone Rail Systems.

The Real-Time Clock Module can be part of any of the Door Module System (DMS) Applications.

1.1.1 Purpose

The purpose of this Software Design Description (SDD) is to describe the software components and implementation details of the Real-Time Clock Module.

The intended audience is the Vapor Stone Rail Systems software development team and maintenance team members of the transit authority and the car builder.

1.1.2 Scope

The Real-Time Clock Module is a software module of the Door Module System (DMS) Application.

1.1.3 Definitions and Acronyms

1.1.3.1 Definitions

Boot Loader	Software starting the Door Control Module application.
Car Network	Data bus and optionally trainlines connecting the Door Control Module with the rest of the train (others Door Control Modules in the car and others cars).
Communication Gateway	A specific processor based electronic module running a Communication Gateway Module Application. It acts as gateway between the Door Control Module Application and the Car Network/Portable Test Application.
Door Module System	A term that is used for generically identifying a processor based electronic module running any software within any Door Control Unit.
Door Control Unit	The final unit that is used to control one or more door panels. This unit is normally made of one or two door control modules running the Door Control Module Application and possibly one Communication Gateway Module Application.
Door Control Module	A specific processor based electronic module running a Door Control Module Application. It is used for controlling the door motor.
HW_COM	Communication interface used for communication between Portable Test Application and Door Control Module application, that can be USB, serial or Ethernet.

1.1.3.2 Event Level Definitions

Event:	Term used to globally identify an Info, an Exception or a Failure.
Info:	Info is an occurrence of an Informative condition monitored by the system, which is neither an exception nor a failure.
Exception:	An exception is an occurrence of a condition that is outside of the specified parameters and affecting an outside Line Replaceable Unit

(LRU).

Failure: A failure is an occurrence of a condition that is outside of the specified parameters and affecting this Line Replaceable Unit. A failure normally requires maintenance or replacement to restore affected unit to its normal operation conditions.

1.1.3.3 Acronyms

CDRL	Contract Data Requirements List
CPU	Central Processing Unit
DCM	Door Control Module
DCU	Door Control Unit
DMS	Door Module System
ECD	Engineering Change Document
HW	Hardware
IEEE	Institute of Electrical and Electronics Engineers
LRU	Line Replaceable Unit
PTA	Portable Test Interface
RTC	Real-Time Clock
SCI	Software Configuration Items
SDD	Software Design Description
SPI	Serial Peripheral Interface
SRAM	Static Random Access Memory
SRS	Software Requirements Specification
Std	Standard
VSRS	Vapor Stone Rail Systems

1.1.4 Document Organization

The Software Design Description (SDD) document describes the architectural software design of the DMS Application Real-Time Clock Module. This SDD is divided into 3 sections and the following specific information is provided within each one:

- Section 1, Introduction, states the purpose, identifies the scope, gives definitions, abbreviations and an overview of the SDD.
- Section 2, References, lists the factors that affect the product and its requirements.
- Section 3, Software design summarizes followings:
    - Decomposition Description: describes functions of software entities.
    - Dependency Description: gives a brief description of the dependency between each entity.
    - Interface Description: describes the interaction between entities.
    - Detailed Design: details the functions implemented by each software entity, presents and explains tasks performed by each software entity in functional conditions.

2 REFERENCES

This section lists the number, title, revision, date, and source of all documents referenced in this SDD.

Standards

IEEE Std 1016-1998	IEEE Recommended Practice for Software Design Descriptions, September 23rd 1998, IEEE, www.ieee.org
IEEE Std 1558-2005	IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems, March 23rd 2005, IEEE, www.ieee.org

Vapor Stone Rail Systems Documents

S17-07	Software Design Description Template, Rev. 0, 2002-01-15 complemented with requirements from IEEE Std 1558.

3 SOFTWARE DESIGN

3.1 Decomposition Description

3.1.1 Module Decomposition

3.1.1.1 Real-Time Clock Module Description

The Real-Time Clock Module is a single entity providing timing functions for all other modules.

3.1.2 Concurrent Process Decomposition

3.1.2.1 Real-Time Clock Task Description

The module shall be linked with the final executable and do not use process or threading. The module functions shall be designed to support multi-threading.

3.1.3 Data Decomposition

The Real-Time Clock module receives time value from PTA and Car Network Interface modules. It sends time value to Fault Logger and Snapshot modules.

The Application Time value *ulTime* is an unsigned 32 bits variable that is incremented by the internal clock of the processor every second. The time value is in seconds since January 1st 1970 and can be stored until year 2106. This variable is saved in the SRAM.

3.2 Dependency Description

3.2.1 Inter-Module Dependencies

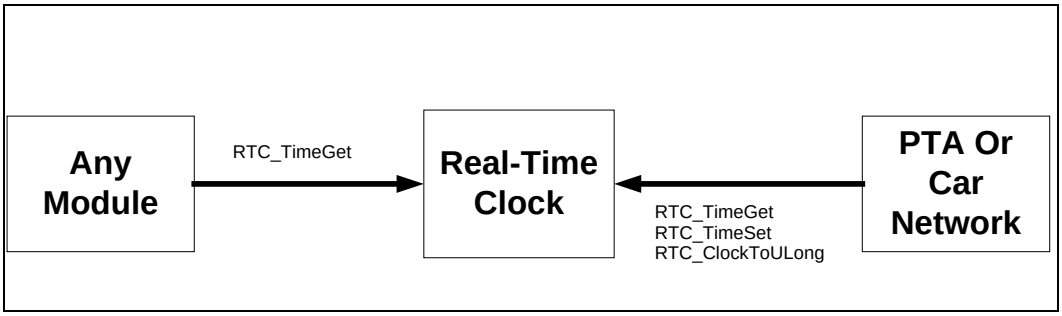

Figure 3-1: Inter-Module Dependencies Diagram

3.2.1.1 Real-Time Clock Module

Figure 3-1 shows the Inter-Module Dependencies Diagram of the Real-Time Clock Module.

3.2.2 Inter-Process Dependencies

Not applicable.

3.2.3 Data Dependencies

Not applicable.

3.3 Interface Description

This section describes the functions of the Real-Time Clock Module. The function name, a description of the function, input and output parameters required by the function, and the name of the module that called the function are provided for each function of each module.

3.3.1 Real-Time Clock Module functions

The Real-Time Clock module implements the following functions (sorted alphabetically):

Function Name	unsigned long RTC_ClockToULong(unsigned int uiYear, unsigned char ucMonth, unsigned char ucDate, unsigned char ucHour, unsigned char ucMinute, unsigned char ucSecond)
Description	Converts date and time value to unsigned long integer value
Input	unsigned int uiYear, the full year unsigned char ucMonth, month (1-12) unsigned char ucDate, date (1-31) unsigned char ucHour, hour (0-23) unsigned char ucMinute, minute (0-60) unsigned char ucSecond, second (0-60)
Output	None
Return	Timestamp in seconds since January 1st 1970

Function Name	unsigned long RTC_TimeGet(void)
Description	Gets the Real-Time Clock value used by the application.
Input	None
Output	None
Return	Timestamp in seconds since January 1st 1970

Function Name	void RTC_TimeSet (unsigned long ulTimeRcv)
Description	Sets the Real-Time Clock variable used by the application.
Input	ulTimeRcv = Timestamp in seconds since January 1st 1970
Output	None
Return	None

3.4 Detailed Design

3.4.1 Real-Time Clock Module Design

3.4.1.1 RTC_ClockToULong

This function validates and converts the received date from the RTC_TimeSet function to an internal value.
This function is too simple to provide an algorithm/flowchart.

3.4.1.1.1 RTC_TimeGet

This function gets the current Application time *ulTime*.
This function is too simple to provide an algorithm/flowchart.

3.4.1.1.2 RTC_TimeSet

This function sets the current Application time *ulTime*.
This function is too simple to provide an algorithm/flowchart.

3.4.2 Data Entities Detailed Design

Not applicable.